3 A cup contains 0.22 kg of water at a temperature 5.0 °C.

A bucket contains 0.50 kg of ice at a temperature of −8.0 °C.

Data:
specific heat capacity of water: $c_{water} = 4.18\ J\ g^{-1}\ °C^{-1}$
specific heat capacity of ice: $c_{ice} = 2.09\ J\ g^{-1}\ °C^{-1}$
latent heat of fusion of ice: $L_f = 334\ J\ g^{-1}$

(a) Determine, for the water in the cup,

(i) the heat released if its temperature decreases to 0 °C;

heat released = ... J [1]

(ii) the heat released by the 0 °C water if it freezes to 0 °C ice;

heat released = ... J [1]

(b) Determine, for the ice in the bucket, the heat absorbed by the ice if its temperature increases to 0 °C.

heat absorbed = ... J [1]

(c) A cup of water, of mass 0.22 kg and temperature 5.0 °C, is poured into a bucket containing ice, of mass 0.50 kg and temperature −8.0 °C.

State and explain the final temperature of the system.

Ignore any heat loss to the surroundings.

final temperature = ... °C

Explanation: .....

.....

.....

..... [2]

[Total: 5]